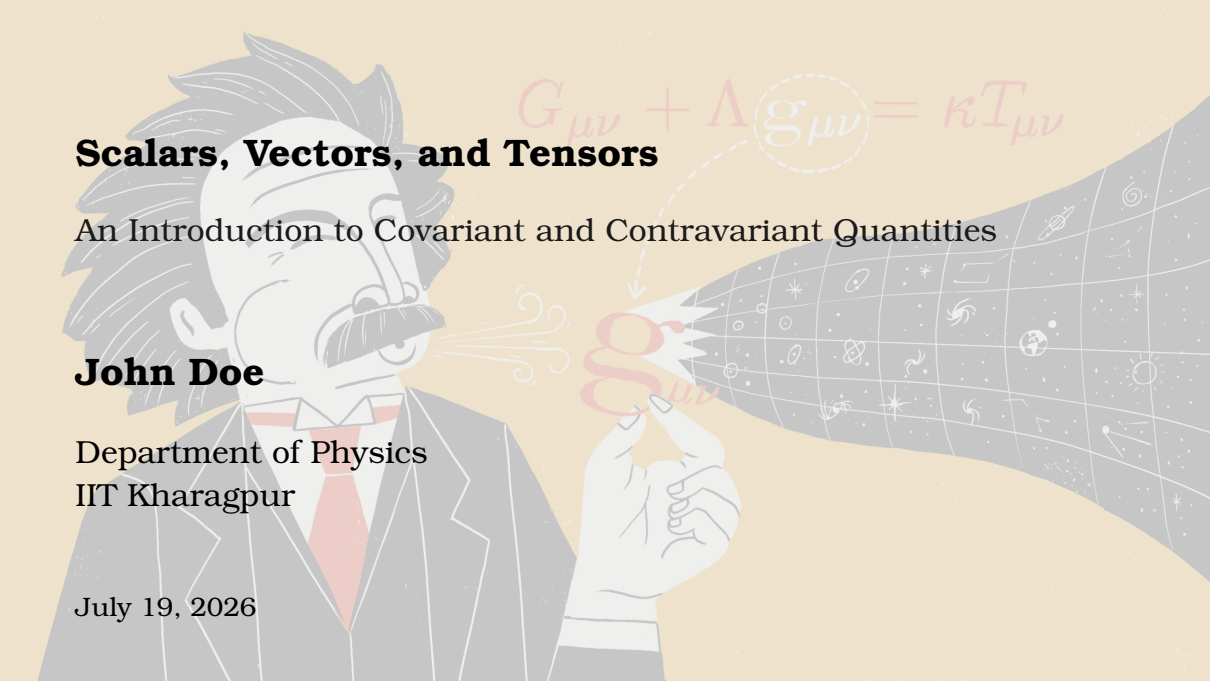


Scalars, Vectors, and Tensors



An Introduction to Covariant and Contravariant Quantities

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OUTLINE

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Motivation

WHY DO WE NEED TENSORS?

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Physical laws should be independent of the coordinate system used to describe them.

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Examples of physical quantities:

- ▶ Temperature
- ▶ Velocity
- ▶ Electric field
- ▶ Stress
- ▶ Space-time metric

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Goal

Develop a mathematical language that works in every coordinate system.

Scalars

SCALARS

Definition

A scalar is a quantity that has magnitude only.

Examples

- ▶ Mass
- ▶ Temperature
- ▶ Electric charge
- ▶ Time
- ▶ Density

Under a coordinate transformation,

$$x^i \longrightarrow x'^i,$$

the scalar remains unchanged,

$$\boxed{\phi' = \phi.}$$

Vectors

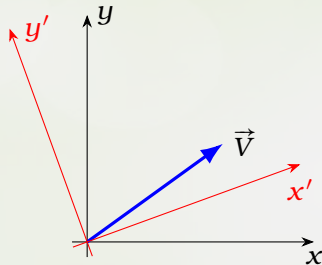
VECTORS

Definition

A vector possesses both magnitude and direction.

Examples

- ▶ Velocity
- ▶ Force
- ▶ Momentum
- ▶ Displacement



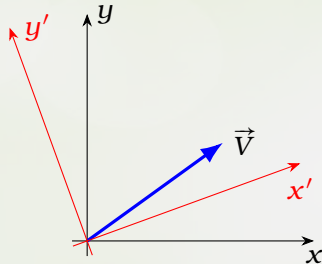
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Only the **components** of a vector change when the coordinate system changes.

Coordinate Systems

COORDINATE TRANSFORMATION

Suppose the coordinates change as

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The geometric vector does not change.

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Examples

- ▶ Cartesian coordinates
- ▶ Polar coordinates
- ▶ Spherical coordinates

Contravariant Vectors

CONTRAVARIANT VECTORS

A contravariant vector has components written with an upper index,

$$V^i.$$

It's transformation law is

$$V'^i = \frac{\partial x'^i}{\partial x^j} V^j.$$

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Examples

- ▶ Position vector
- ▶ Velocity
- ▶ Displacement

Remember

Contravariant components transform with the Jacobian.

Covariant Vectors

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A covariant vector has components

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Examples

- ▶ Gradient ($\partial_i \phi$)
- ▶ Differential forms (dx_i)

Comparison

COVARIANT VS CONTRAVARIANT

Property	Contravariant	Covariant
Index	V^i	V_i
Transformation	$\frac{\partial x^i}{\partial x'^j}$	$\frac{\partial x'^j}{\partial x^i}$
Examples	Velocity	Gradient

Observation

One transforms with the Jacobian, the other with its inverse.

Dot Product

WHY TWO DIFFERENT TYPES?

Consider

$$A_i B^i.$$

After changing coordinates,

$$A'_i B'^i = \left(\frac{\partial x'^j}{\partial x^i} A_j \right) \left(\frac{\partial x^i}{\partial x'^k} B^k \right).$$

The Jacobians cancel,

$$A'_i B'^i = A_i B^i.$$

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Conclusion

The scalar product is independent of the coordinate system.

Tensors

TENSORS

A tensor may possess several upper and lower indices.

Examples

$$T_{ij}, \quad T_j^i, \quad R^\rho_{\sigma\mu\nu}.$$

Object	Rank	Example
Scalar	0	mass m
Vector	1	position $\vec{r}$
Metric Tensor	2	$g_{\mu\nu}$
Riemann Tensor	4	$R^\rho_{\sigma\mu\nu}$

HOW A TENSOR TRANSFORMS

A rank 2 covariant tensor (say Electromagnetic-Field Tensor) transforms as :

$$F'_{ij} = \frac{\partial x^m}{\partial x'^i} \frac{\partial x^n}{\partial x'^j} F_{mn} \quad (1)$$

While a contravariant one changes as :

$$F'^{ij} = \frac{\partial x'^i}{\partial x^m} \frac{\partial x'^j}{\partial x^n} F^{mn} \quad (2)$$

HOW A TENSOR TRANSFORMS

For mixed tensor of rank 2, the transformation law is :

$$F^i_j = \frac{\partial x'^i}{\partial x^m} \frac{\partial x^n}{\partial x'^j} F^m_n \quad (3)$$

Shortcut

- ▶ For each upper index, write a term of this form $\frac{\partial x'^i}{\partial x^m}$.
- ▶ For each lower index, write a term of this form $\frac{\partial x^n}{\partial x'^j}$.
- ▶ *Voilà!* You're now a tensor transformation expert.

Metric Tensor

RAISING AND LOWERING INDICES

The metric tensor connects vectors and covectors.

$$V_i = g_{ij} V^j$$

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Interpretation

The metric acts as a bridge between covariant and contravariant components.

Summary

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SUMMARY

- ▶ Scalars remain unchanged under coordinate transformations.
- ▶ Vectors are geometric objects.
- ▶ Components depend on the coordinate system.
- ▶ Contravariant vectors have upper indices.
- ▶ Covariant vectors have lower indices.
- ▶ Tensors provide a coordinate-independent description of physics.



Thank You

Questions?